Milestone 2 Progress Report

Real-Time Human Activity Classification Using CNN and LSTM/Transformer

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1. Where I Am

At this stage of the project, I have successfully implemented a preliminary deep learning model using a stack of **ConvLSTM2D** layers for human activity classification in videos. While the architecture is functioning as expected, I am currently in the *optimization and experimentation phase* as I work on improving performance and reducing computational cost.

2. What I've Done So Far

- Implemented a **ConvLSTM-based model** that captures both spatial and temporal features.
- Constructed a full pipeline including `MaxPooling3D`, `TimeDistributed`, `Flatten`, and `Dense` layers.
- Verified architecture and compilation, with output layers compatible for 4-class classification.
- Identified areas for improvement and began exploring alternative architectures.

Model Architecture Summary

Layer	Type	Output Shape	Parameters
conv_lst_m2d_8	ConvLSTM2D	(None, 20, 62, 62, 4)	1024
max_pooling3d_8	MaxPooling3D	(None, 20, 31, 31, 4)	0
time_distributed_6	TimeDistributed	(None, 20, 31, 31, 4)	0
conv_lst_m2d_9	ConvLSTM2D	(None, 20, 29, 29, 8)	3488
max_pooling3d_9	MaxPooling3D	(None, 20, 15, 15, 8)	0
time_distributed_7	TimeDistributed	(None, 20, 15, 15, 8)	0
conv_lst_m2d_10	ConvLSTM2D	(None, 20, 13, 13, 14)	11144
max_pooling3d_10	MaxPooling3D	(None, 20, 7, 7, 14)	0
time_distributed_8	TimeDistributed	(None, 20, 7, 7, 14)	0
conv_lst_m2d_11	ConvLSTM2D	(None, 20, 5, 5, 16)	17344
max_pooling3d_11	MaxPooling3D	(None, 20, 3, 3, 16)	0
flatten_2	Flatten	(None, 2880)	0
dense_2	Dense	(None, 4)	11524

Total Parameters: 44,524
Trainable Parameters: 44,524
Non-trainable Parameters: 0

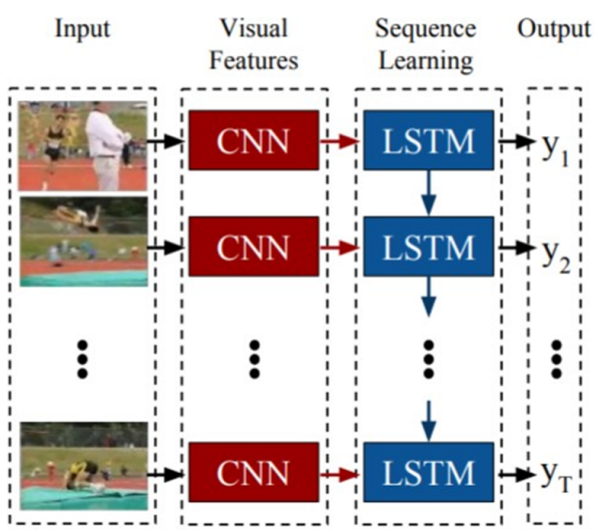

Figure 1: Approach

3. What I've Learned

Conceptual Insights

- **ConvLSTM2D** is effective at modeling both spatial and temporal information simultaneously, making it ideal for end-to-end spatiotemporal modeling.

- It operates on 3D input (Height $\times$ Width $\times$ Channels), unlike standard LSTM, which only works with 1D input.
- The model complexity grows quickly, making it computationally expensive for long sequences or high-resolution video.

Misunderstandings Realized

- I underestimated the GPU/TPU requirements for training ConvLSTM architectures.
- Initially assumed ConvLSTM would outperform CNN+LSTM hybrids, but realized that modular approaches are more flexible and resource-efficient.

4. What I'm Still Having Difficulty With

- **Training performance:** Due to resource constraints, training has been slower than expected.
- **Architectural tuning:** Adjusting filters, pooling sizes, and layer depth is challenging without efficient experimentation.
- **Real-time inference:** ConvLSTM may not be optimal for deployment due to its computational cost.

5. What I Still Plan or Would Like to Do

I plan to explore a more efficient and modular approach using the **Long-term Recurrent Convolutional Network (LRCN)**. This architecture separates spatial and temporal learning into different modules.

Future Directions

1. **LRCN (End-to-End):** Combine CNN and LSTM in a single model. CNN layers extract features from frames, LSTM layers model sequence dynamics.
2. **CNN + LSTM (Two-Stage):** Use a pretrained CNN to extract features, then pass them to an LSTM. This modular approach enables faster training and use of transfer learning.

These alternatives are more feasible for real-time systems and make it easier to experiment with different configurations.

6. How My Plans Have Changed from the First Proposal

- I originally proposed a hybrid CNN+LSTM/Transformer model. While the core idea remains, I've shifted away from ConvLSTM as the primary solution.
- LRCN and modular CNN+LSTM pipelines now seem more scalable, efficient, and better suited for resource-constrained environments.
- Transformer-based models may still be explored in the future, but likely after a stable LRCN implementation is completed.

Conclusion

This milestone has laid a strong foundation for understanding the trade-offs between end-to-end spatiotemporal models and modular architectures. While ConvLSTM offers theoretical advantages, practical constraints have guided me toward exploring LRCN and CNN+LSTM pipelines for better performance, efficiency, and real-time capability.